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New approaches to eliciting pro repertoire manipulation: the co

To date, the fundamental principles of vaccination have remained unchanged. The overriding concept for each vaccine has been the establishment of protective immunity largely due to antigen-specific T cell expansion, facilitating subsequent proliferation and differentiation of CD8 cytotoxic effector T cells and CD4 helper T cells capable of producing antiviral cytokines and chemokines (Fig. 1A).

CD4 T cells activate B cells to generate neutralizing antibodies, offering protection against viral attachment/ translocation or bacterial toxins, etc. [4,5].

Microorganisms including HIV and malaria, among others, may alter their antigenic proteins through rapid mutagenesis, thereby hindering cytotoxic T lymphocyte (CTL)-based immunity, exploiting holes in the T cell repertoire, and/or misdirecting both cellular and humoral responses away from key cell-binding receptors to pathogen components which cannot provide epitopes for neutralizing antibodies [10-13].

Finding ways to increase the pool of mature, primed T cells that are able to fend off disease is a goal for future vaccine development.

In this respect, a novel strategy for vaccine design, termed "thymic vaccination" has been considered to alter the T cell antigen receptor repertoire centrally via altered peptide ligands (APL).

APL derived from infectious agents or tumor antigens, with low affinity to the TCR could, in principle, mediate positive selection and export of specific T cells from the thymus [14].

As such, these APL might be candidates for manipulating the thymic repertoire in vivo, controlling the generation of naive T cells and hence, subsequent memory development within the peripheral lymphoid compartment.

Through repertoire manipulation, it should be possible to sculpt the specificity and diversity of diseasefighting cells.

Note how engendering T cells with anti-viral specificity requires administration of an APL, a weaker affinity ligand for a given TCR to encourage maturation and emigration from the thymus.

Expanding T cell repertoires has enormous potential in aiding the organism's fight against infections or in affording tumor immunity.

This strategy is principally different from conventional "peripheral" vaccination, which leads to proliferation of pre-existing mature T cells but does not alter the repertoire through creation of T lymphocytes with new T cell specificities.

Thymic vaccination, by contrast, will alter the thymic repertoire to create desired T cell specificities.

T cells bearing a highly diverse T cell receptor (TCR) repertoire develop in the thymus from stem cells originating in the hemopoietic tissues [15-17].

On entering the thymus through the cortico-medullary junction, these cells migrate to the subcapsular epithelium and undergo a complex differentiation process in the thymic cortex and then in the medulla, involving proliferation, expression of accessory molecules, rearrangement of TCR genes and selection of the TCR repertoire (reviewed in [18,19]).

T cell development does not occur autonomously but requires signals from non-hematopoietic stromal cells including various types of thymic epithelial cells (TECs) which show profound phenotypic differences between cortex and medulla.

Thymic nurse cells, expressing high levels of MHC class I and II molecules and also containing antigen processing machinery, are involved in thymocyte selection, mediated by peptide/MHC (pMHC) ligands.

Pools of self-peptides bound to MHC molecules control both positive and negative selection (reviewed in [20]).

Thymocytes that carry TCRs having low-affinity interactions with MHC-bound self-peptides are positively selected, and are exported into the pool of mature Fgineuerrea l2scheme of thymocyte selection and emigration to th General scheme of thymocyte selection and emigration to th expressing pre-TCR undergo divisions and become TCR⁺ presented by class I and II MHC molecules expressed on thy high affinity to pMHC undergo apoptosis, while those bound restricted CD8⁺ SP or class II-restricted CD4⁺ SP cells.

In contrast, thymocytes bearing those TCRs that recognize self-peptides with "high" affinity are eliminated primarily upon interaction with dendritic cells [19,21].

The avidity of pMHC/TCR interactions plays a major role in T cell recognition [14].

Crystal structure analyses have revealed fine details about peptide conformation inside the peptide binding groove of MHC molecules and the amino acid residues interacting with the TCR V and V domains including their CDR3 loops [22-25].

Peptide analogs of antigenic peptides with substitutes at amino acid residues, APL, have been shown to generate qualitatively different T cell responses compared with those produced by the antigenic peptides themselves [26].

CD4-CD8⁻ (DN) cells D4⁺CD8⁺ (DP) at which stage they interact with self-peptides ic stromal cells.

Those thymocytes whose TCRs interact with o pMHC with low affinity mature to become MHC class I mature thymocytes then emigrate to the periphery aided by difnucleoprotein peptide of the influenza virus NP₃₆₆₋₃₇₉ in the context of H-2D^b, a peptide antagonist mediates positive selection in FTOC [35,36], whereas the cognate peptide itself leads to deletion of DP thymocytes [37].

Design and initial characterization of APL derived from the viral epitopes gp₃₃₋₄₁ and VSV8 To experimentally test the concept of thymic vaccination, we have designed variants of gp₃₃₋₄₁ and VSV8 peptides with substitutions at residues interacting with the TCR aimed at reducing TCR-pMHC affinity via diminution of the number of atomic contacts between the peptide and the TCR.

In the case of gp₃₃₋₄₁ cognate peptide, amino acids at the peptide positions p4 (Tyr, Y) and p6 (Phe, F) were modified to Ser (S) and Ala (A), respectively, based on the crystal structure of the gp₃₃₋₄₁/H-2D^b complex showing exposure of the side chains of these amino acid residues to the solvent and hence, TCR accessibility [38,39].

No change was made in the peptide anchor residues that occupy the binding pockets of H-2D^b, thus ensuring proper peptide presentation in the context of MHC.

In the other less extreme peptide variant of gp₃₃₋₄₁, Ala (A) at p7 was substituted with Glu (E).

For the VSV8 peptide, the weak L4 agonist with the substitution of Leu (L) for Val (V) at the p4 peptide residue has been employed. The crystal structure of the N15 TCR-VSV8/K^b complex as well as the space-filling models of K^b in complex with VSV8 and L4 peptides are shown in Fig.

The centrally positioned p4 peptide residue, whose atoms are shown in green in the space-filling model, faces up to the solvent and interacts with the N15 TCR.

The binding of the APL to the MHC class I molecules using RMA-S cells confirmed that amino acid substitutions at peptide residues interacting with the TCR did not affect peptide binding and, by extension, peptide presentation to T cells.

A series of experiments was next performed to evaluate the functional potential of the APL to stimulate peripheral T cells in mice injected with the variant peptides.

Results of both proliferation and cytokine secretion assays using mature T cells from P14Rag2^{-/-} lymph node and spleen showed a response to the high TCR affinity cognate peptide gp₃₃₋₄₁, but not to the Y4S/ F6A variant peptide.

In support of this observation, tetramers of H-2D^b in complex with the Y4S/F6A peptide did not bind to SP CD8 thymocytes from P14 Rag2^{-/-} mice at any tetramer dilution, as judged by immunofluorescence analysis, whereas high fluorescence intensity staining was detected using tetramers of H-2D^b in complex with the gp₃₃₋₄₁ peptide.

Effect of APL administration on thymocyte development in vivo Injection of the cognate viral peptides gp₃₃₋₄₁ and VSV8 leads to negative selection of P14- and N15TCR-bearing thymocytes, respectively, due to relatively high affinity pMHC/TCR interactions [31,34].

Surprisingly, injection of P14 Rag2^{-/-} mice with the Y4S/F6A peptide mutant resulted in a significant increase in the total number of thymocytes as well as the DP thymocyte subpopulation, while the A7E variant had no effect on thymocyte counts.

The unusual increase in the number of DP thymocytes following exposure to Y4S/F6A peptide

was not due to cellular proliferation and attendant DNA synthesis as examined by in vivo BrdU incorporation assay.

Rather, Y4S/F6A peptide administration prevented apoptosis as confirmed by staining of thymocytes with anti-Annexin V mAb.

To more directly test this hypothesis, we injected P14 Rag2^{-/-} mice with mixtures of the negatively selecting cognate peptide gp₃₃₋₄₁ plus the Y4S/F6A APL.

Increasing the amount of Y4S/F6A peptide in the injection mixture resulted in a higher number of total and DP thymocytes.

Thus, we infer that the Y4S/F6A variant may compete with other endogenous negatively-selecting peptides for binding to H-2D^b molecules expressed on thymic stroma either by binding to "empty" surface MHC class I molecules or, perhaps, by a cross-presentation mechanism [41].

Y4S/F6A and L4 peptides mediate positive selection and emigration of thymocytes in irradiation chimeras. The numerically small population of antigen-specific recent thymic emigrants (RTE) makes thymic selection/ F127guucrtu 3al basis of APL design Structural basis of APL design.

For this purpose, lineage-minus BM precursors of P14 or N15- TCR transgenic Rag2^{-/-} mice (donor) were injected into irradiated congenic B6 mice (recipient) and the development of donor-type cells was monitored weekly by immunofluorescence staining and multicolor FACS analysis [40].

The figure was rendered in MOLSin in blue, K^b in magenta and 2M in red. Note the VSV8 peptide space-filling models of K^b in complex with VSV8 peptide and with surface [59] in magenta with peptide in CPK format and p4 resiboth DP and SP CD8 donor thymocytes were highest, consistent with positive selection.

In contrast, injection of either gp₃₃₋₄₁ or VSV8 cognate viral peptides into irradiation chimeras led to thymocyte depletion by negative selection.

Analysis of the peripheral lymphoid organs in these chimeras by triple color immunofluorescence with anti-CD45.2, anti-CD8 and anti-TCR-specific mAbs showed the greatest number of donor-type CD45.2⁺CD8⁺V 2⁺ T cells in the lymph nodes of Y4S/F6A - injected chimeras (2-3 fold over PBS-injected control mice), suggesting that donor-type thymocytes expressing the P14 TCR had developed in the presence of Y4S/F6A, matured and emigrated to the lymph nodes.

A similar increase in the donor cell numbers were observed up to 9 weeks after injection of Y4S/F6A peptide.

In this case, higher CD8⁺V 5.2⁺ N15 TCR transgene donor-type T cell numbers were observed both in lymph nodes and spleens.

The functional analysis of donor-type CD8⁺ lymph node T cells in irradiation chimeras injected with the positively selecting Y4S/F6A or L4 peptides showed approximately two-fold higher proliferation levels in response to the cognate peptides gp₃₃₋₄₁ and VSV8, respectively, in vitro, compared to cells from PBS control-injected chimeric mice, reflecting the two-fold difference in the number of donortype CD8⁺ T cells in lymph nodes of chimeras injected with the APL.

However, importantly, these mature donortype T cells did not proliferate in response to Y4S/F6A or L4 variant peptides in vivo.

Injection of the viral peptides and their APLs in vivo led to reduction of CFSE⁺ staining in the case of gp₃₃₋₄₁ and VSV8, suggestive of proliferation and/or activation-induced cell death (AICD).

In contrast, no change in CFSE⁺ staining was observed upon injection of Y4S/F6A or L4 peptides, implying that these APLs do not facilitate T cell expansion per se.

In sum, the cognate peptide ligands gp₃₃₋₄₁ and VSV8 which interact with the TCR with relatively high affinity compared to their respective APL, induce activation of peripheral T cells, whereas peptide variants Y4S/F6A and L4, which bind TCR with low affinity and mediate positive selection, do not stimulate mature T cell divisions.

These results suggest that although the low affinity pMHC/TCR interactions are insufficient to trigger cell divisions in mature cells, differentiation of immature thymocytes nevertheless follows.

Affinity measurements support the idea that positively selecting peptide ligand affinities are lower than those of negatively selecting ligands for TCRs, but additionally linked to their MHC binding/stability properties [42].

Certainly the 10,000 fold weaker functional stimulation of N15-bearing T cells by L4 versus VSV8 peptide is in line with the view [14].

Collectively, our data show that cognate peptides can be modified at key TCR recognition positions to create variants that result in selection, directly or indirectly, of desired TCR specificities at the level of thymic development.

This exogenous peptide administration offers a potential of expanding repertoire generation in vivo in a manner useful to the organism.

Whether these peptidespecific T cells generate stronger defense mechanisms to fight viral infection or tumors in normal, non-transgenic mice remains to be investigated. The magnitude of the APL-driven increase in thymocytes and subsequent egress is only 2-3 fold, however.

In this respect, exploring peptide-based means of enhancing differentiation of thymocytes bearing desired TCRs together with the modulation of mechanisms controlling thymocyte emigration to the periphery would be of a great importance.

Although not discussed further here, thymic vaccination followed by conventional cognate antigen immunization may be the best way to insure a robust memory T cell response.

As thymocytes progress through the developmental stages, they migrate from the cortico-medullary junction, the site of entry of T cell progenitors from the BM, to the subcapsular region of the thymus, then to the cortex and to the medulla [18,19].

The processes that regulate trafficking of lymphoid precursors to and within the thymus, and that mediate emigration of mature T cells from the thymus to the periphery remain poorly understood.

Several mediators, including chemokine receptors [45], adhesion molecules [46], extracellular matrix proteins [47], neuroendocrine factors [48] and G-protein coupled receptors (GPCR) [49] have been shown to regulate thymocyte export (Fig. 2).

Recently, a role for the early activation marker CD69, transiently expressed on activated mature T cells and on thymocytes undergoing positive selection, in controlling thymocyte export, has also been suggested [50].

Cellular mechanisms involved in thymocyte egress are discussed in the following sections.

[45,51,52]) Chemokines are basic polypeptides of about 100 amino acids, usually containing four Cys residues linked by disulphide bonds, which are produced by certain thymic stroma cells and are abundantly expressed in the thymus.

Specifically, thymic epithelial, medullary epithelial and dendritic cells have been shown to secrete various chemokines.

Growing evidence suggests that chemokines and their receptors, expressed differentially on thymocytes during discrete maturational stages, control homing of T cell progenitors to the thymus, their intrathymic migration, and exit to the periphery.

Chemokines stimulate responding cells by activating pertussis toxin-sensitive G_i protein-coupled seven-transmembrane receptors (GPCR), leading to activation of intracellular secondary mediators which control directional cell migration.

In particular, CCR7 has been demonstrated to mediate homing of naïve T cells to peripheral lymphoid organs via ligands CCL19 and CCL21.

Extracellular matrix proteins (reviewed in [47]) Extracellular matrix (ECM) proteins laminin and fibronectin are produced by TECs, fibroblasts and MHC class II⁺ macrophages in the thymus.

ECM proteins form molecular bridges between thymocytes and the thymic microenvironment, mediating adhesion of thymocytes via their ECM receptors VLA-4, -5 and -6, and their disassembly from the cell complexes.

The expression of S1P₁ on T cells controls their exit from the thymus and entry into the blood, and, thus, has a central role in regulating the numbers of peripheral T-cells [55].

Interestingly, S1P₁ knock-out mice show a block in the egress of mature T-cells into the periphery.

The regulated expression of S1P₁ receptor levels, which is increased in mature SP thymocytes and peripheral T cells, may control responsiveness to the high levels of sphingosine 1-phosphate in the blood, which selectively induces mature T-cell migration to the periphery.

Recently, S1P₁ receptors have been implicated in lymphocyte trafficking and homing based on studies using FTY720, a potent immunosuppressive agent, which is an agonist ligand for S1P₁, 3,4,5 receptors blocking egress of T cells from the thymus.

Studies of thymocyte egress mechanisms through the S1P receptor pathway may aid in facilitating emigration from the thymus to the periphery and provide additional means of enriching the mature T cell pool with desired specificities.

Thymic vaccination might offer promising clinical applications as a way of immunization against these infectious diseases and cancers, enabling "designer" thymic development to produce suitable and long-lasting protective T cell immune specificities.

In a converse role, deletion of unwanted T cell specificities in the case of autoimmunity by agonist administration early in life could be considered. Assuming thymic vaccination proves clinically viable, immune responses against invariant components of infectious agents, such as HIV and malaria, which otherwise utilize their intrinsic mutational capacity to evade human immune recognition, can be targeted. Design of novel vaccines based on the thymic vaccination approaches will benefit from the information gained in the recently completed Human Genome Project, particularly as genetic polymorphism associated with high risk of developing certain diseases later in life including cancers, infectious disease susceptibility and autoimmunity are uncovered. For example, the ability to manipulate the T cell repertoire to elicit anti-tumor responses early in life may prevent clinical disease evolution later.

This approach differs substantially from conventional vaccination since it aims to shape T cell responses through thymic repertoire manipulation, exposing developing thymocytes to positively selecting APL derived from infectious agents or tumors.

Increasing emigration of antigen-specific T cells from the thymus to the periphery is a challenging goal.

APL, altered peptide ligand; BM, bone marrow; CMJ, cortico-medullary junction; DC, dendritic cells; DN, double negative; DP, double positive; GPCR, G-protein-coupled receptors; HEV, high endothelial venule; MMP, matrix metalloproteinases; RTE, recent thymic emigrants; SP, single positive; S1P, sphingosine 1-phosphate; TCR, T cell receptor; TEC, thymic epithelial cells.